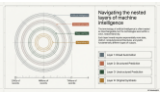
Use notebook LLM to convert Class room file Agenda--Day1 to ppt or deck:

Answer:

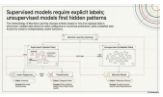


Artificial Intelligence mimics human problem-solving through pattern recognition
AI is broader definition. It encompasses different ways of thinking to create smart solutions. Machine Learning is a specific subset of AI that focuses on learning from data to make predictions.
Inputs
Engine
Output
Sensors
Systems
Traffic Jams
Weather Patterns
Behavior Patterns
AI
Machine Learning
Algorithm Trained Between Sports Brands and Retailers to Predict Trends
Data Collection
Collect Data
Feature Engineering
Extract Important Features

Machine Learning relies on structured datasets to predict outcomes
Machine Learning is a subset of AI that focuses on learning from data to make predictions.
Unlike AI, which aims to mimic human intelligence broadly, ML specifically uses algorithms to analyze data and make decisions.
Collect Data
Clean Data
Train Model
Make Predictions
Deploy Model
PREDICTION

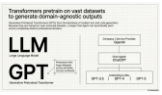
The predictive pipeline converts raw inputs into clinical insights
Training a model requires moving from raw, restricted data collection through rigorous cleaning and structure to a mathematical model. Which is accessible only to clinical staff and protected by strict access controls.
Training Pipeline
Raw Collection
Data Cleaning
Model Selection
Feature Engineering
Hyperparameter Tuning
Cross-Validation
Best Model Selected
Model Deployed

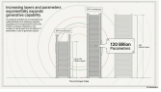
Models prove their accuracy against withheld validation data
The model's job is to learn patterns from historical data. Then apply those patterns to new, unseen data. The only way to know if it works is to hold back a portion of the data and test the model against it.
The 80/20 Data Split
80% Training Data
Raw Dataset
Full clinical history
20% Validation Data
Withheld from training
Model Learning Phase
Validation Phase
Model Accuracy Report
Sensitivity, Specificity, Confusion Matrix
Ready for deployment


Supervised models require explicit labels; unsupervised models find hidden patterns
The distinguishing of Machine Learning strategy evolves based on the first decision made: whether the training data provided to the algorithm includes pre-assigned correct answers (labels) or whether the algorithm must independently discover structure within raw, unlabeled data.
Machine Learning Strategy
Supervised Learning Path
Unsupervised Learning Path
Labeled Training Data
Algorithm Learns Mapping
Predictions
Unlabeled Training Data
Algorithm Identifies Patterns
Groups, Associations, Anomalies
Model Outputs Predictions on New Data
Model Outputs Groups, Associations, Or Anomalies

Deep Learning introduces neural networks to handle trillion-record scale
Deep Learning employs layered neural networks to learn hierarchical representations from raw data, automatically discovering features that human engineers would miss. Its computational requirements scale with data volume, making it dependent on distributed cloud infrastructure.
Operating Scale
Millions To Trillions Of Records
Training Large Models
Hours To Weeks
Cloud Infrastructure Required

Generative AI abandons prediction to synthesize entirely new content
In the newest wave of deep learning (by Generative AI, instead of analyzing data to predict outcomes, it analyzes it to produce entirely new content, or creative output.)
PARADIGM SHIFT: NOT PREDICTIVE
Gen AI Engine
Image
Text
Audio
Video
Code and Software


Transformers pretrain on vast datasets to generate domain-agnostic outputs
Transformers are a type of neural network architecture that transforms or changes input sequences into output sequences.
LLM
Large Language Model
GPT
Generative Pre-trained Transformer
Company Open-Ended Generative Agents
Microsoft ChatGPT
Microsoft Copilot
GPT-3.5
GPT-4
GPT-4o
GPT-5


Increasing layers and parameters , enhance the model generative capability
Large language models are characterized by their vast number of parameters, which underpin their capacity to process and generate complex text. The scale of these models, measured in billions of parameters, allows them to capture nuanced patterns in language and perform a wide range of tasks with high accuracy.
Transformer Layer
Feed Forward Layer
Attention Layer
Input Embedding Layer

The interconnected ecosystem of artificial intelligence

An analytical framework for data science, machine learning, and generative AI, mapping the core concepts, methods, and relationships that define modern artificial intelligence. From computer vision to generative AI, each layer builds fundamentally upon the previous layer.

	Machine Learning	Deep Learning	Generative AI
Core Inputs	Tabular Data, Features & Records	Unstructured Data (Images, Text, Audio)	Massive Pre-trained Datasets
Primary Output	Predictions	Predictions	Creations
Key Architecture	Linear Regression, Decision Trees, SVM	Artificial Neural Networks, CNNs, RNNs	Transformers & Diffusion Models
Training Focus	Feature Engineering, Data Prep	Network Design, Hyperparameter Tuning	Prompt Engineering, Alignment, RLHF